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Sleep Efficiency Classification Using Machine Learning

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Background

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Sleep quality is influenced by everyday lifestyle behaviors such as physical activity, diet, and substance use. In this project, we explore how these factors relate to sleep health using the NHANES Sleep Data dataset.

- Data

We chose the NHANES dataset, which contains daily records of individuals' behavours and sleep-related measures.

This includes:

- Sleep metrics (e.g., duration, timing, efficiency)
- Physical activity and sedentary behavior
- Dietary intake
- Substance use

- Problem

We aim to identify which lifestyle factors are most strongly associated with sleep efficiency. Specifically, we want to understand which daily behaviors are linked to better or worse sleep quality, in order to identify potential targets for improving sleep.

Datasets

- Dataset

We use the NHANES Sleep Data dataset, which contains daily records of individuals' sleep and lifestyle behaviors. It combines sleep measurements with daily activity, diet, and health-related behaviors collected from participants.

- Features

We selected features related to daily lifestyle and behavior including:

- Physical activity levels
- Sleep duration and sleep timing variables
- Dietary intake
- Substance use indicators

These features are used to explain variations in sleep quality.

- Model Output

The model is designed to predict whether an individual had good or poor sleep based on daily lifestyle and behavioral features.

- Samples with sleep efficiency >= 0.85 are labeled as 1 (good sleep)
- Samples with sleep efficiency < 0.85 are labeled as 0 (poor sleep)
- 66% of dataset is good sleep, 33% is bad sleep

Sleep Efficiency = (time spent in bed - time spent awake)/(time spent in bed)

Related work

Freeman et al., “Monitoring sleep duration, timing, and continuity among US youth and adults in NHANES using actigraphy”

- Used NHANES actigraphy* data to measure sleep duration, timing, and continuity
- Created the main dataset and methodology for our project.
- Used wrist-worn actigraphy to estimate sleep duration/timing/continuity.
- Results showed differences in sleep patterns across age groups, weekdays and weekends, and demographic groups

Su et al., 2022 — “Epidemiology of accelerometer-based sleep parameters in US school-aged children and adults: NHANES 2011–2014”

- Used our same dataset to estimate objective sleep parameters across U.S population.
- Focused on descriptive analysis instead of ML prediction
- Found that sleep duration followed a parabola shape with age, reaching its lowest point at age 40 while sleep efficiency declined with age.

Cai et al., 2025 — “Association between accelerometer-measured physical activity volume and sleep duration in older adults”

- Studies if physical activity predicts sleep duration
- Compared multiple models including logistic regression, random forest, XGBoost, etc.
- Observed that physical activity was associated with good sleep duration

Methods

- **Data Pre-processing**
  - Remove unnecessary/redundant columns.
  - Convert variables to type float.
  - Classify data points by using sleep_efficiency >= 0.85.
  - Split data 70/15/15 for training, testing, and validation.
- **Models**
  - Linear Perceptron
  - Logistic Regression
  - Random Forest
  - Neural Network

Experimental setups

Features

- Actigraphy circadian rhythm metrics: L5/M5/L10/M10 values (least and most active periods), and their timing (L5TIME_num, M5TIME_num)
- Sleep timing: sleep period time (dur_spt_min), bracelet wear time (dur_day_min), day of week, wake up time
- Signal quality: percent of day device was not worn (nonwear_perc_day)
- Sleep-period acceleration (ACC_spt_sleep_mg)
- Circadian ratio = M5VALUE / L5VALUE

Target

Good sleep quality if sleep efficiency >= 0.85, bad sleep quality otherwise

Different Model Setups

- Pandas, numpy, scikit-learn, PyTorch (nn only)
- NN, RF used 12 features, L.R used 10 features, L.P used 10

Hyperparameters

Random Forest Setup

Parameter	Value	Reasoning
n_estimators	200	Enough trees to stabilize variance
max_depth	10	Prevents overfitting on data
min_samples_leaf	5	Smooths leaf predictions

Linear Perceptron Setup

Parameter	Value	Reasoning
Learning Rate	0.1	Large enough to learn and small enough to avoid instability
Epochs	100	Enough to converge on linear separable dataset

Hyperparameters

Neural Network Setup

Parameter	Value	Reasoning
Architecture	$12 \rightarrow 64 \rightarrow 32 \rightarrow 1$	Two hidden layers, enough capacity for 12 features
Activation Function	ReLU	Avoids vanishing gradient
Dropout	0.3	Regularization after hidden layers
Batch Normalization	Occurs after linear layers	Stabilizes the training
Epochs	100	Avoids overfitting
Batch Size	256	Suitable for dataset

Logistic Regression Setup

Parameter	Value	Reasoning
Max Iterations	1000	Safe default for finding optimal weights
Regularization	L2	Prevents model from over-relying on single feature
C	1.0	Neutral value that shrinks weights but doesn't ignore regularization

Experimental results

Results	Linear Perceptron	Logistic Regression	Random Forest	Neural Network
Accuracy	0.516	0.5658	0.5950	0.6220
Precision	0.6669	0.7225	0.7424	0.7446
Recall	0.4991	0.5568	0.5931	0.6515
F-1	0.5709	0.6289	0.6594	0.6949

Linear Regression → Logistic Regression → Random Forest → Neural Network

- Neural networks and random forest predicted sleep quality most consistently/accurately
- Models struggled with poor class more than good class (due to split)
- Gap between simpler and complex models suggest that data is nonlinear

Model Weights

Logistic Regression

Feature	Coefficient
% sleep device not worn	+0.54
sleeponset	+0.28
M10VALUE	+0.21
wakeup	-0.30
% day device not worn	-0.46

Random Forest

Feature	Coefficient
Sleep period duration	0.197
L5VALUE	0.148
Wrist Acceleration	0.131
Circadian Ratio (peak/rest act)	0.108
Device Wear Time	0.093

Model Weights

Linear Perceptron

Feature	Coefficient
% of day device worn	+0.5446
Wake up time	-0.5076
M5TIME	-0.4898
M5VALUE	0.4716
M10VALUE	-0.3584

Neural Network

Feature	Coefficient
L5VALUE	0.7713
Wrist Acceleration during sleep	0.6821
Time in bed	0.4081
L10VALUE	0.1412
M10VALUE	0.1025

Conclusion

Key Findings

- Successfully classified sleep efficiency using actigraphy-based lifestyle features
- Sleep efficiency is strongly influenced by lifestyle factors, but relationships are nonlinear
- Circadian rhythm and activity pattern metrics were among the most predictive features

Challenges

- Class imbalance (more "good sleep" samples than "poor sleep") led to reduced performance on poor sleep prediction
- Limited to actigraphy-based features, which may not capture all behavioral or physiological influences on sleep

References

Freeman, Joshua R et al. “Monitoring sleep duration, timing, and continuity among US youth and adults in NHANES using actigraphy.” Sleep health vol. 12,1 (2026): 104-112. doi:10.1016/j.sleh.2025.10.008, Accessed

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